SubTrack: Design Justification

Android subscription tracker (Flutter + Dart) · Mobile Programming milestone

1. Intended users and design goals

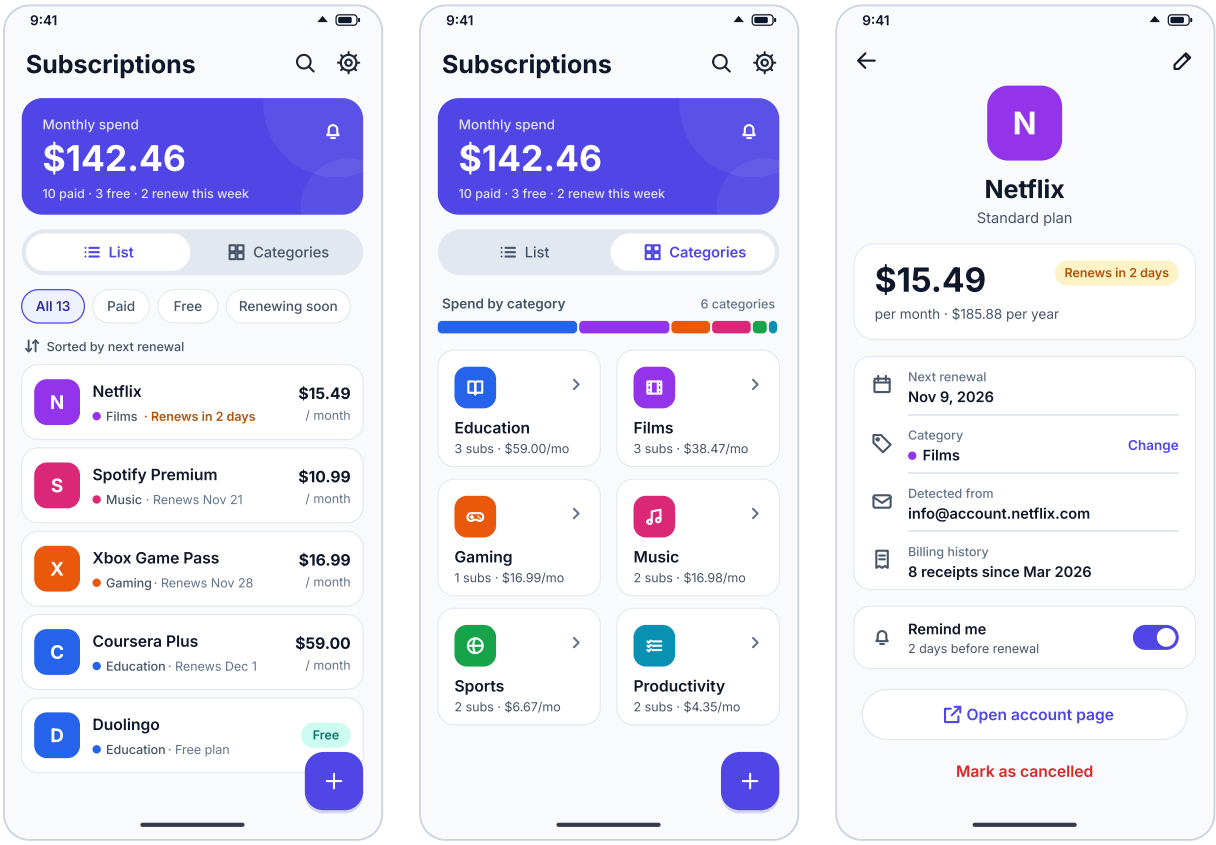
SubTrack is for **students and young professionals** who pay for many small digital services (streaming, music, online courses, gym and sports apps, games) and lose track of what renews when. They are comfortable with Android, check the app in short sessions, and are **cautious about giving an app access to their email**. Three goals came from this: (1) **clarity**: the total monthly spend and the next renewals should be readable in under five seconds; (2) **trust**: the app must explain exactly what it reads before asking for Gmail access; (3) **flexibility**: users can switch between a flat list and a view grouped by category (education, sports, films, and so on).

2. Colour palette

The palette is a calm, neutral base (slate greys on near-white) with **one strong brand colour, indigo**. Indigo suggests reliability and finance without the "bank-blue" coldness, and it stays distinct from every category colour, so the user always knows what is tappable. Status colours carry fixed meanings: **teal = free**, **amber = renews soon**, **red = destructive**. All text/background pairs below were checked against WCAG 2.1 and meet **AA (4.5:1)** for normal text.

Role	Hex	Used for	Contrast
Primary (Indigo 600)	`#4F46E5`	Primary buttons, links, selected tabs, FAB	on `#FFFFFF`: 6.3:1
Primary dark	`#3730A3`	Text on light primary (selected chips)	on `#EEF2FF`: 8.9:1
Text	`#0F172A`	Headings, names, prices	on `#FFFFFF`: 17.9:1
Text secondary	`#475569`	Dates, captions, metadata	on `#FFFFFF`: 7.6:1
Teal (Free)	`#0F766E`	"Free" badges, success ticks	on `#CCFBF1`: 4.9:1
Amber (Renews soon)	`#B45309`	Renewal warnings	on `#FEF3C7`: 4.5:1
Red (Destructive)	`#DC2626`	Cancel, delete, sign out	on `#FFFFFF`: 4.8:1

Category colours (used only for small icons, dots and the spend bar, never for body text): Education Films Gaming Music Sports Productivity . They are spread across the hue wheel so neighbouring categories stay distinguishable. They are also always paired with a text label and an icon, so **colour is never the only signal** (important for colour-blind users).



Mockups: list view · category view · subscription detail

3. Typography

Inter was chosen because it is designed for screens, has a large x-height that stays readable at 12 sp, and has **tabular (equal-width) numerals**, so prices such as $15.49 and $59.00 line up in a column. It is open-source (SIL OFL), available through the `google_fonts` package, and pairs naturally with Android's Roboto. The type scale follows Material 3, with a limited set of sizes so the hierarchy stays obvious:

Style	Size / weight	Example use
Display	32 sp · Bold	Monthly total "$142.46", app name
Headline	24 sp · Bold	Screen titles ("Subscriptions")
Title	15–20 sp · SemiBold	Service names, app-bar titles, buttons
Body / Label	14 sp Regular · 12 sp Medium	Descriptions · dates, categories, captions

4. Layout

- **8 pt grid, 16 dp side margins, 16–20 dp card radius**: a consistent rhythm that matches Material 3.
- **Summary first**: the indigo "Monthly spend" card sits at the top, because the total is the question users open the app to answer.
- **Segmented control (List | Categories)** switches the view in place instead of hiding it in a menu, which answers the requirement that users can choose a different interface.
- **Cards with a fixed structure**: logo at left, name and category in the middle, price at right. The eye scans the price column straight down.
- **Thumb zone**: the main action (+ Add) is a bottom-right FAB, and primary buttons sit in the lower third of the screen.
- **Progressive disclosure**: the list shows only name, category, renewal and price. Billing history, the sender address and reminders appear on the detail screen.

5. Accessibility

- All text meets WCAG AA contrast (see table). Price and status never rely on colour alone (e.g. the "Free" text badge, the "Renews in 2 days" wording).
- Touch targets are at least **48 × 48 dp** (list rows are 72 dp, buttons 48–54 dp).
- Layouts use flexible rows so **system font scaling up to 200 %** wraps text instead of clipping it. Every icon button has a TalkBack `Semantics` label.
- Follows the system **dark theme**. The palette has dark equivalents that keep the same roles.

6. How the visual design supports trust

Because the app asks for Gmail access, the permission screen is designed as reassurance rather than a legal wall: a shield icon, three short cards ("What we read / What we never do / Where your data stays") and the exact scope name (`gmail.readonly`). The secondary "Not now, add manually" option is always visible, so users never feel forced to grant access. Calm colours, rounded shapes and plain language reduce the feeling of being "scanned", while the live list on the scanning screen shows the user what the app is actually finding.